

Program: Diploma in Hotel Management and Catering Technology	
Course Code: 3472	Course Title: Principles of Nutrition
Semester: 3	Credits: 3
Course Category: Program core	
Periods per week: 3 (L:3 T:0 P: 0)	Periods per semester: 45

Course Objectives:

- Develop an understanding of nutrition, its definition, and its importance in human health
- Understand the six classes of nutrients: Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water
- Have an idea of Food Preparation Techniques for Nutrient Retention and Special Diets
- Food Safety and Hygiene related to Nutrition

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food	Food Production I & II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop an understanding of nutrition, its definition, and its importance in human health	6	Understanding
CO2	Understand the six classes of nutrients: Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water	18	Understanding
CO3	Have an idea of Food Preparation Techniques for Nutrient Retention and Special Diets	9	Understanding
CO4	Explain the instrumentation and working of equipment used for food analysis	12	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2		3					
CO3			2				
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop an understanding of nutrition, its definition, and its importance in human health		
M1.01	Definition of nutrition and understanding nutrition	1	Understanding
M1.02	Role in Nutrition on health	1	Understanding
M1.03	Classification and role of each nutrient in human health	4	Understanding
Contents: Definition, Role of nutrition in health, classification and role of each in human health			
CO2	Understand the six classes of nutrients: Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water		
M2.01	Carbohydrates Types of carbohydrates: Simple vs. complex Food sources of carbohydrates	3	Understanding
M2.02	Proteins Complete vs. incomplete proteins Sources of protein in different diets (plant-based vs. animal-based)	3	Understanding
M2.03	Fats and Lipids Types of fats: Saturated, unsaturated, and trans fats Cholesterol and its impact on health Sources of healthy and unhealthy fats	3	Understanding

M2.04	Vitamins Water-soluble vs. fat-soluble vitamins Sources of key vitamins: A, B-complex, C, D, E, and K Deficiencies and toxicity of vitamins	3	Understanding
M2.05	Minerals Major minerals and Trace minerals Sources of essential minerals The role of minerals in metabolism, growth, and development	3	Understanding
M2.06	Water – The Essential Nutrient The importance of water in the body Daily water requirements	2	Understanding
	Series Test –I	1	
Contents: Types, sources , importance and health effects of Carbohydrates, Proteins, Fats, Vitamins, Minerals, and Water			
CO3	Have an idea of Food Preparation Techniques for Nutrient Retention and Special Diets		
M3.01	Cooking methods that preserve nutrients (steaming, grilling, baking vs. frying).	3	Understanding
M3.02	Special Diets	3	Understanding
M3.03	Portion control and serving sizes for guests	3	Understanding
Contents: Cooking methods to preserve nutrients, Special Diets and Serving Sizes			
CO4	Food Safety and Hygiene related to Nutrition		
M4.01	Importance of food hygiene in preserving nutritional quality	3	Understanding
M4.02	Safe food handling, storage, and preparation techniques	4	Understanding
M4.03	Preventing contamination and foodborne illnesses Best practices for maintaining a clean and safe kitchen in a hotel environment	4	Understanding
	Series Test–II	1	
Contents: Importance of food hygiene in preserving nutrition, safe food handling, storage and preparation techniques, Preventing health issues and Best Practices			

Text / Reference

T/R	Book Title/Author
R1	"Nutrition and Dietetics for the Hospitality Industry" by J. J. Givens
R2	Food Science - B. Srilakshmi
R3	Essentials of Nutrition and Dietetics - Swaminathan
R4	Dietetics - B. Srilakshmi

Online Resources

Sl.No	WebsiteLink
1	Nutrition.org, The Academy of Nutrition and Dietetics (eatright.org), Food Safety and Standards Authority of India (FSSAI).
2	www.nin.res.in (National Institute of Nutrition)
3	www.who.int (World Health Organization - Nutrition)
4	www.fao.org (Food and Agriculture Organization)